Pick, Price, Portfolio

An AI-augmented security analysis of Hindustan Unilever, Sun Pharma and ICICI Bank, and the 15-year SIP we would build from them

Investor: us, TAPMI BBA(H) Semester V students starting a ₹10,000 monthly SIP. **Data:** Yahoo Finance, Screener, BSE filings, RBI, MOSPI, Damodaran, Google News; model numbers frozen on 13 Oct 2026.

Our recommendation

Own the three stocks as a satellite (HUL 31%, Sun Pharma 19%, ICICI Bank 50%) around a Nifty 50 index core. Our call on all three is **Hold at these weights and add only through the monthly SIP**. We do not claim any of them is mispriced: our tests find no reliable way to beat these prices (§3) and no alpha we can tell from zero (§4). Our valuations show instead what each price requires you to believe (§2). A Balanced investor (A = 4) puts 44% of savings into the three stocks and the rest into T-bills. A ₹10,000 SIP rising 10% a year has a median 15-year value of ₹64.8 lakh from ₹38.1 lakh invested.

1. Company selection: three sectors, three different drivers

We worked top-down (BKM Ch 17). India's real GDP grew 7.8% in Q1 FY27, led by investment (GFCF +11.9%) rather than household consumption (+7.1%); the repo rate is 5.25% and the 91-day T-bill 5.52% (MOSPI, RBI). We looked for stocks with (i) a different primary macro driver, (ii) a five-year daily-return correlation of 0.5 or less with each other, (iii) Nifty 50 liquidity and at least six quarters of call transcripts, and (iv) a business that a standard valuation model fits.

Company	Sector and main driver	Role in the portfolio	Beta	CAGR 2016–26	Sharpe
Hindustan Unilever	FMCG: consumption, input costs	Defensive anchor: lowest beta, in the weakest sector (FMCG 10.6% below its 200-day average)	0.60	9.6%	0.28
Sun Pharma	Pharma: chronic therapy, US specialty, USD/INR	Defensive growth: demand ignores the cycle and US sales earn dollars	0.65	8.7%	0.24
ICICI Bank	Private bank: credit growth, rates, asset quality	Growth engine of an investment-led cycle: loans +19.6% a year	1.29	19.1%	0.55
Nifty 50 (est. TR)	Benchmark	Price index + 1.22% a year estimated dividends	1.00	12.0%	0.45

Table 1. The three picks. Risk and return from daily data, 2016 to 13 Oct 2026.

Pairwise correlations are 0.16–0.24 over 2016–2026 (0.15–0.20 over five years), so the three diversify each other. We rejected Media, the strongest sector (11.2% above its 200-day average): at A = 4 its certainty equivalents are −8% to −38%, below the T-bill, and buying it now would be recency bias. Infosys, our first choice, was replaced by Sun Pharma, which keeps the dollar hedge at a lower beta (0.65 vs 0.82); HDFC Bank's 2023 merger and ITC's 2025 demerger break their histories. **Honesty note:** we chose Sun Pharma partly after its strong run, which flatters our 2023–26 back-test (§4).

Street check, 3 Nov. Analysts share our view of three separate drivers but rank the stocks almost opposite to our models: Buy ratings are 100% for ICICI, 85% for Sun Pharma and 69% for HUL. None of our four tests claims alpha, so this does not undo the selection. One test has weakened: Sun Pharma's $11.75bn all-cash purchase of Organon, due to close in Q4 FY27, turns net cash into about 2.3× net debt/EBITDA, so its 2016–26 beta understates future risk (roughly 14% higher on a rough Hamada estimate) and our model fits it less well.

2. Traditional analysis: what these prices require

The cost of equity is the CAPM required return, r = rf + β × ERP, with rf = 5.52% and Damodaran's India ERP of 7.08%. HUL pays out 67% of its profit, so we value it with a two-stage dividend discount model. Sun Pharma (payout 34%) and ICICI Bank, whose borrowings are raw material, use a two-stage residual-income model, equivalent to a justified P/B of (ROE − g) / (r − g). Peer multiples give a second view (P/E for HUL and Sun Pharma, ROE-adjusted P/B for ICICI), and the blended value averages the two. Every working is a live formula in valuation.xlsx, with r × g sensitivity grids.

Stock	Price	r	Model value	Peer value	Blended	Blended vs price	Implied growth	Analyst target
HUL	₹1,862	9.8%	₹1,079	₹2,105	₹1,592	−14%	7.4%	₹2,396 (+29%)
Sun Pharma	₹1,760	10.1%	₹748	₹1,702	₹1,225	−30%	8.6%	₹2,174 (+24%)
ICICI Bank	₹1,351	14.7%	₹575	₹1,139	₹857	−37%	13.9%	₹1,747 (+29%)

Table 2. Intrinsic value vs market price, 13 Oct 2026. Implied growth: the perpetual growth rate at which our model equals the price.

Stock	P/E	P/B	ROE	Payout	Beta	Max drawdown	Daily VaR 95%
HUL	39.1	9.0	21.9%*	67%	0.60	−36%	2.0%
Sun Pharma	33.4	5.1	13.7%	34%	0.65	−62%	2.5%
ICICI Bank	16.8	2.7	15.0%	14%	1.29	−48%	2.5%

*Table 3. Key ratios. FY26 fundamentals extracted from the annual reports (§3); market ratios 2016–2026. *HUL ROE excludes a one-off gain of about ₹3,900 Cr.*

Every price is above our blended value, but we do not read that as the market being wrong. Reversing the models gives what each price requires: perpetual growth of 7.4% (HUL), 8.6% (Sun Pharma), 13.9% (ICICI Bank), or a discount rate well below our CAPM rate (Table 4). HUL's "fairly valued" label also rested on a ±15% band: −14.5% at the freeze, −17.7% at the 3 Nov price. So we make no valuation-based Buy.

Stock	Our r	Our value	r on 5-yr β	Value at 5-yr β	r needed for price	r needed for target	Analysts	Buy share	Mean target	Since 13 Oct
HUL	9.8%	₹1,079	9.6%	₹1,122	7.9%	7.6%	39	69%	₹2,256 (+17%)	−5.8%
Sun Pharma	10.1%	₹748	9.5%	₹866	7.6%	7.2%	33	85%	₹2,136 (+22%)	−1.7%
ICICI Bank	14.7%	₹575	12.3%	₹831	9.9%	9.1%	42	100%	₹1,769 (+26%)	+1.3%

Table 4. What each price requires, and what the Street says. Prices of 3 Nov 2026, read that morning; Yahoo consensus. The 13 Oct model inputs are not changed.

Why the Street sits higher. For ICICI, we can walk from our value to the price one assumption at a time: ₹575 on our model; ₹831 with the five-year beta (0.96, not 1.29); ₹1,096 after ₹265 for the subsidiaries (Motilal Oswal's sum of the parts); ₹1,213 with the Street's ROE (17%, then 16%); against a price of ₹1,402 and a mean target of ₹1,769. The gap is inputs, not facts. HUL's price needs a long growth runway: 10% for 10 years, 8% for 10 years, then 5.5% gives ₹1,673. We keep the full-sample beta as our base case, because switching after seeing the Street's answer would be tuning; the five-year beta would make P 39/21/40 and A = 4 38% in P. The same 1.29 beta makes ICICI both the largest weight and the widest valuation gap, which is consistent once the portfolio is read as "take prices as efficient" and the valuation as "what must be true".

3. AI-augmented analysis: five tasks, each checked

We used AI where it reads or labels text faster than we can, and checked each task against an independent benchmark. Numbers are computed in Python, never generated by a model. The prompts (P1–P5) and their outputs are saved in docs/ai_prompts.md and data/ai/.

Task (brief)	What the AI did	Checked against	Result
Ratio extraction from PDFs	Claude read the FY26 annual reports and returned 47 line items as JSON, each with a page reference	Yahoo Finance and Screener	44 of 47 within 1%; 3 definition differences (e.g. Yahoo's bank cash includes money at call); 0 errors
Sentiment: calls and news	FinBERT scored 2,428 headlines (12 months) and 9,133 call sentences, split into management and analysts	Claude's call tone (P4): same sign on 3 of 3 latest calls	Management sounds more positive than analysts on every call: gap 0.28 (HUL), 0.08 (Sun Pharma), 0.14 (ICICI Bank)
Behavioural bias, news framing	Claude tagged 120 headlines for gain/loss framing, anchoring on targets, authority, herding, hype, fear and recency	FinBERT labels (70% agreement)	41% carry a bias cue, mostly authority and herding; 14% are not about the company; Sun Pharma coverage 60% gain-framed
Forecasting; is TA possible?	50/200-day trend and RSI rules after 0.25% costs; earnings event study (CAR[−1,+1]); consensus EPS and targets	Buy-and-hold; earnings surprises	0 of 8 rules beat buy-and-hold on Sharpe; +1.2% after beats, −1.6% after misses (72 events)
Risk heatmaps	Correlation, 120-day rolling correlation, risk contribution, stress tests	Four sell-offs since 2020	ICICI supplies 57% of risk at 50% weight; P fell less than the Nifty in 4 of 4

Table 5. The five AI tasks in the brief, with their checks.

What we learned. AI made it practical to read every page and headline, but two findings argue for scepticism. Managements talk up their results, so we discount call tone. Sun Pharma's coverage is 60% gain-framed and only 3% loss-framed, the one-sided optimism our valuation warns about.

Form	Test	Result	Reading
Weak	8 trading rules after a 0.25% round-trip cost	0 of 8 beat buy-and-hold on Sharpe	Past prices did not pay
Weak	Lag-1 autocorrelation and variance ratios at 5, 20 and 60 days, with robust z	Largest |z| is 1.9, below the 2 that would flag a pattern	No detectable pattern
Semi-strong	Reaction to earnings, days −1 to +1, over 72 results	Beats +1.22% (t 2.2), misses −1.57% (t -2.2); gap +2.79% [+1.04%, +4.54%]	News is priced quickly
Semi-strong	Drift after the news: beat minus miss, days +2 to +20 and +2 to +60	−0.53% [−2.80%, +1.74%]; +1.64% [−2.21%, +5.49%]	No drift detected, but the ranges are wide
Strong	Not tested	It needs inside information we do not have	No claim made

Table 6. Market-efficiency evidence, 2016–2026. Abnormal returns from a market model, with 95% confidence intervals. Reaction grows with surprise size (r = 0.32, p = 0.009, excluding surprises of 30% or more).

Efficiency. Our evidence fits weak-form and semi-strong efficiency, with one limit: 72 events can miss a small drift. Harshita found about 6% over 64 days in a long-short Nifty 500 portfolio, a size our range does not rule out. This is why our project does not hunt for alpha; mosaic research here checks the thesis rather than finding mispricing, and SPIVA's finding that 73% of Indian large-cap funds lagged over 10 years supports the index core.

Fear and greed. CNN's index is American; the Indian equivalent, Tickertape's Market Mood Index, read 62.7 (Greed) on 3 Nov, a single hand reading with no history. For the stocks we score four price signals against each stock's own last three years: HUL 39, Sun Pharma 37, ICICI 58 (13 Oct: 17, 18, 36; 0 = fear, 100 = greed). It is built from prices, so it describes mood rather than measuring it independently. No stock-level test reaches $p < 0.05$, and FinBERT tone says nothing about the next five days ($|r| \leq 0.08$). At market level, a higher India VIX was followed by higher Nifty returns ($r = 0.24$, $p = 0.005$; without COVID, $r = 0.20$, $p = 0.024$), but one quintile carries it, and that may be a risk premium rather than a mispricing. Sentiment here describes the crowd; it does not time it.

4. Portfolio and the long-term plan

We maximise the Sharpe ratio with 10–60% in each stock, using the 2016–26 covariance matrix. With historical mean returns, the optimiser pushes ICICI to the 60% cap (Sharpe 0.58), which is hindsight: weights fitted on 2016–22 earned 3.2% a year in 2023–26, against 7.4% for the Nifty. We therefore use forward-looking CAPM expected returns, which give P = HUL 31%, Sun Pharma 19%, ICICI 50%: E(r) 12.3%, σ 19.7%, Sharpe 0.34. Fitted the same way on 2016–22, they earned 7.9% a year out of sample.

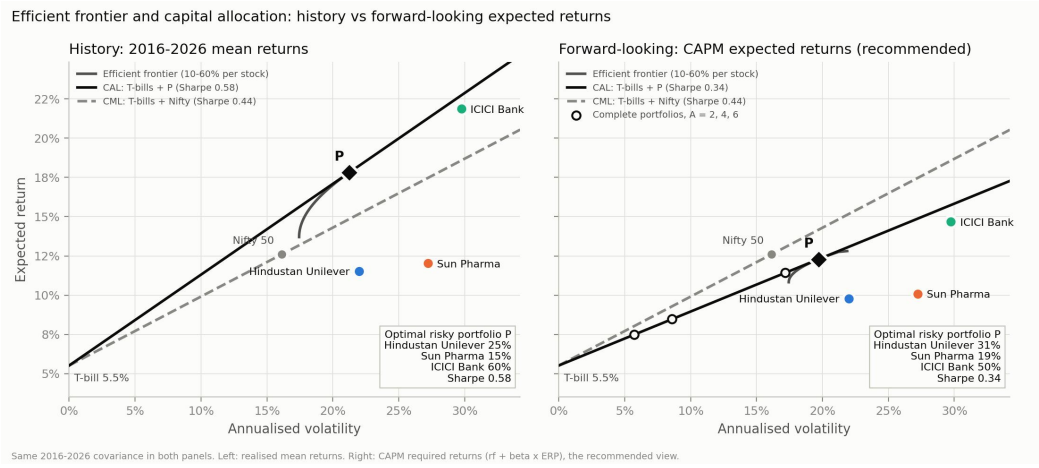


Figure 1. Efficient frontier, CAL (T-bills + P) and CML (T-bills + Nifty) under historical and forward-looking expected returns.

P's Sharpe ratio (0.34) is below the market's (0.44): three stocks cannot match the index's diversification, so our CAL lies under the CML. That is why the long-term plan is core-satellite. Each investor's share in P is $y^* = (E[r_P] - r_f) / (A\sigma_P^2)$, capped at 100% (no borrowing); the rest goes to T-bills.

Profile	A	Share in P	E(rc)	σc	Certainty equiv.	1-yr 95% VaR on ₹1 lakh	P(loss), 1 yr	15-yr SIP median	P(≥ ₹75 lakh)
Aggressive	2	87%	11.4%	17.2%	8.5%	₹16,138 / ₹15,984	28% / 27%	₹69.5 lakh	36% / 29%
Balanced	4	44%	8.5%	8.6%	7.0%	₹5,582 / ₹5,465	17% / 16%	₹64.8 lakh	19% / 11%
Conservative	6	29%	7.5%	5.7%	6.5%	₹1,940 / ₹1,822	10% / 9%	₹61.3 lakh	4% / 1%

Table 7. Complete portfolios and their risk. Pairs are parametric (normal) / block-bootstrap Monte Carlo, 10,000 paths each.

Portfolio	CAGR	Rp − Rf	Rp − Rm	σ	β	Sharpe	Treynor	Jensen's α	t(α)	Max DD
2023–26: Forward-looking P	7.9%	+3.0%	+0.5%	12.8%	0.71	0.24	4.3%	+1.2%	0.27	−16%
2023–26: Equal weight	8.7%	+3.8%	+1.2%	12.3%	0.65	0.31	5.8%	+2.1%	0.47	−18%
2023–26: History-based P	3.2%	−1.3%	−3.8%	13.7%	0.65	−0.09	−2.0%	−3.0%	−0.55	−20%
2023–26: Nifty 50	7.4%	+2.5%	–	12.6%	1.00	0.20	2.5%	+0.0%	–	−15%
2016–26: HUL	9.6%	+6.2%	−1.1%	22.0%	0.60	0.28	10.2%	+1.8%	0.32	−36%
2016–26: Sun Pharma	8.7%	+6.7%	−0.6%	27.2%	0.65	0.24	10.3%	+2.0%	0.26	−62%
2016–26: ICICI Bank	19.1%	+16.5%	+9.3%	29.7%	1.29	0.55	12.8%	+7.1%	1.17	−48%
2016–26: Nifty 50 (est. TR)	12.0%	+7.2%	–	16.1%	1.00	0.45	7.2%	+0.0%	–	−38%

Table 8. Performance measures on realised returns. Sharpe = (Rp − Rf) / σ. Treynor = (Rp − Rf) / β. α = (Rp − Rf) − β(Rm − Rf), with Newey-West t-statistics (5 lags). 2023–26 weights fitted on 2016–22.

Did it beat the market? In 2023–26, forward-looking P earned 3.0% a year over T-bills against the Nifty's 2.5%, and beat it on Sharpe (0.24 vs 0.20) and Treynor (4.3% vs 2.5%), but its α of +1.2% has a t-statistic of only 0.27, and no α in the table reaches 2. Equal weights did better and history-based weights lost, so this test shows no weighting skill. Sharpe judges a whole portfolio; Treynor and α judge a piece of a diversified one, which is how the satellite is used. Going forward, CAPM inputs give every asset α = 0 and Treynor = 7.08% (the ERP) by construction, so these measures only mean something on realised returns. Sharpe and Treynor do not change with A; α on your money scales with y: 1.06% at A = 2, 0.53% at A = 4, 0.35% at A = 6.

Building it over the years. Each of us estimates A with a short questionnaire (docs/risk_questionnaire.md) and starts a ₹10,000 SIP that rises 10% a year: half into a Nifty 50 index fund and half into P, with equity gliding to 40% as the goal nears. Our Investment Policy Statement sets the rules. Keep six months of expenses outside the portfolio. Rebalance yearly, with new money first. Review after every results season. Never sell in the month after a 20% fall, and never buy because a stock has just risen. The bootstrap keeps real fat tails and volatility clustering; its VaR is close to the normal model's, so fat tails matter little over one year.

5. Recommendation, risks and limitations

Stock	Call	Range	Price	Why
Hindustan Unilever	HOLD	₹1,592–₹2,396	₹1,862	Lowest beta, kept for defence. Q2 volume +7% but margin down 142 bp; the Street trimmed its mean target 5.8%. The 3 Nov price needs r of 7.9% against our 9.8%
Sun Pharma	HOLD	₹1,225–₹2,174	₹1,760	Residual income says expensive, peers say fair. Organon's debt is not in its beta or in some broker estimates, so we would not add above the SIP until it closes
ICICI Bank	HOLD	₹857–₹1,747	₹1,351	Highest expected return and every analyst's Buy, but 57% of portfolio risk. The 3 Nov price needs r of 9.9% against our 14.7%

Table 9. Calls. The range runs from our blended value to the analysts' mean target; prices are as of 13 Oct. Hold = keep the portfolio weight and add only through the SIP.

Risks. Concentration: ICICI is 50% of the weight but 57% of the risk. Valuation: all three prices require more growth, or a lower discount rate, than our base case. Sun Pharma: Organon's leverage and a proposed cap on trade margins for anti-cancer drugs. Cycle: a credit slowdown would hit ICICI and the Nifty together. In four real sell-offs P fell less than the Nifty every time (Appendix D).

Limitations. One risk-free rate is applied to 2016–26; the Nifty total return is estimated (+1.22% a year); CAPM is a single-factor model and forward returns depend on the ERP; the Monte Carlo assumes stable volatility and correlation; results are pre-tax and before fees; Sun Pharma was chosen after its run. Broker views are mostly second-hand press summaries read on 3 Nov, not the original notes (data/street/street_views.csv marks each one), and the Sun Pharma Q2 figures differ between sources. This is coursework, not investment advice.

APPENDIX · NOT PART OF THE FOUR-PAGE REPORT

A. How to view the dashboard

Link: added once the app is deployed on Streamlit Community Cloud (see README).

- If you see "This app has gone to sleep", click the button; it loads in about a minute. Free apps sleep when idle.
- The data selector opens on Submission, the numbers in this report. Latest is refreshed every weekday evening after the NSE close.
- It opens with a 10-question risk questionnaire that sets your risk aversion A (or skip it with A = 4). The terminal then shows your portfolio in rupees, and every figure (allocation, one-year VaR, the 15-year SIP, stress losses) recomputes when you move A or the amount.
- "Ask the portfolio" shows saved answers; live questions use a free OpenRouter model and cite the file behind each number.

Code and data: GitHub repository (link added once published); a ZIP of the tagged submission is attached.

B. What we submit

File	Contents
SA_Portfolio_Report.pdf	This report
SA_Portfolio_Deck.pptx / .pdf	The 15–20 minute presentation, with speaker notes
valuation.xlsx	DDM, residual income, peer multiples and sensitivity grids, as live formulas
sa-portfolio (GitHub / ZIP)	Python pipeline, tests, dashboard, AI outputs and the frozen data behind every number

C. Data sources and AI steps

Source	Used for
Yahoo Finance (yfinance)	Daily prices 2016–2026, statements, analyst consensus, earnings dates
Screener.in	Ten-year financials; peer multiples
BSE filings, company IR sites	FY26 annual reports and 6–7 quarters of call transcripts per company
RBI; MOSPI	Risk-free rate (91-day T-bill, 5.52%) and repo rate; GDP and demand mix
Damodaran (Jan 2026)	India equity risk premium, 7.08%
Google News RSS; Zerodha Aftermarket Report	12 months of headlines; sector distance from moving averages (12 Oct 2026)
P1, P4, P5: Claude (Anthropic)	Ratio extraction; call aspects and guidance; headline framing and bias tags
P2, P3: FinBERT (ProsusAI/finbert)	Sentence-level tone of calls and headlines
Yahoo Finance, Screener; press summaries of broker notes, read 3 Nov 2026	Consensus, Q2 FY27 results and broker calls (each marked first- or second-hand in data/street/street_views.csv)
Tickertape Market Mood Index; ^INDIAVIX	Market fear and greed: one reading on 3 Nov 2026; India VIX history
Harshita, Singh & Yadav (2018), Theoretical Economics Letters 8(14)	Post-earnings drift exists in Nifty 500 stocks, Q4 2002 to Q3 2017: a long-short SUE portfolio earned about 6% over 64 days
Sharda (2022), review, emerald xjm 19(1)	Evidence on analyst recommendations is mixed; 57.6% of Buy targets were reached within a year (Sayed & Chaklader, 2014)
SPIVA India, via Angel One (Oct 2025)	73% of large-cap funds lagged their benchmark over 10 years, 90% over 5

D. Further figures and tables

Sell-off	HUL	Sun Pharma	ICICI Bank	Portfolio P	Nifty 50
COVID crash (Feb-Mar 2020)	−18.4%	−19.6%	−47.9%	−34.6%	−37.2%
Rate-hike sell-off (Jan-Jun 2022)	−10.9%	−6.3%	−16.2%	−12.3%	−16.1%
Election-result shock (4 Jun 2024)	+6.0%	−1.6%	−7.6%	−2.3%	−5.9%
2026 drawdown (02 Jan 2026 to 13 Oct 2026)	−19.9%	+2.8%	+0.5%	−5.1%	−13.7%

Table A1. Stress tests: price change through four real sell-offs. P fell less than the Nifty in 4 of 4.

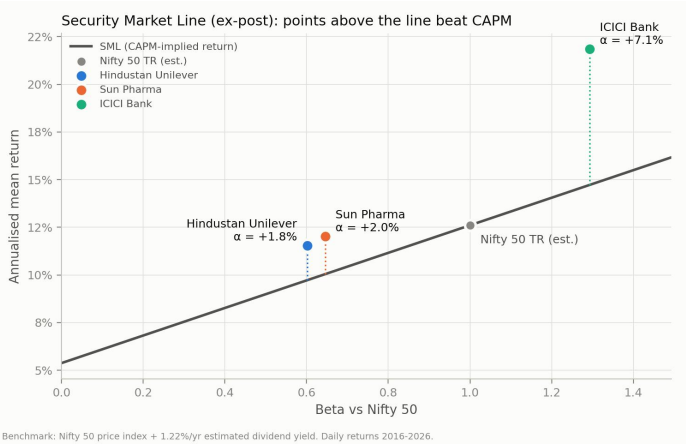


Figure A1. Ex-post Security Market Line, 2016–2026. The distance above the line is Jensen's α (Table 8).

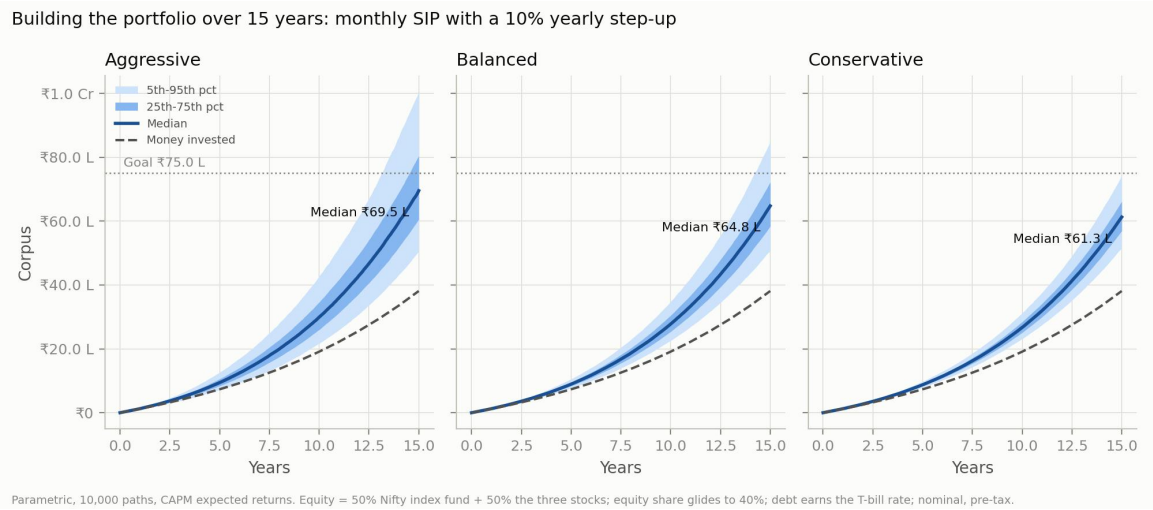


Figure A2. Corpus from a ₹10,000 monthly SIP rising 10% a year: median, 25th–75th and 5th–95th percentiles, against money invested.

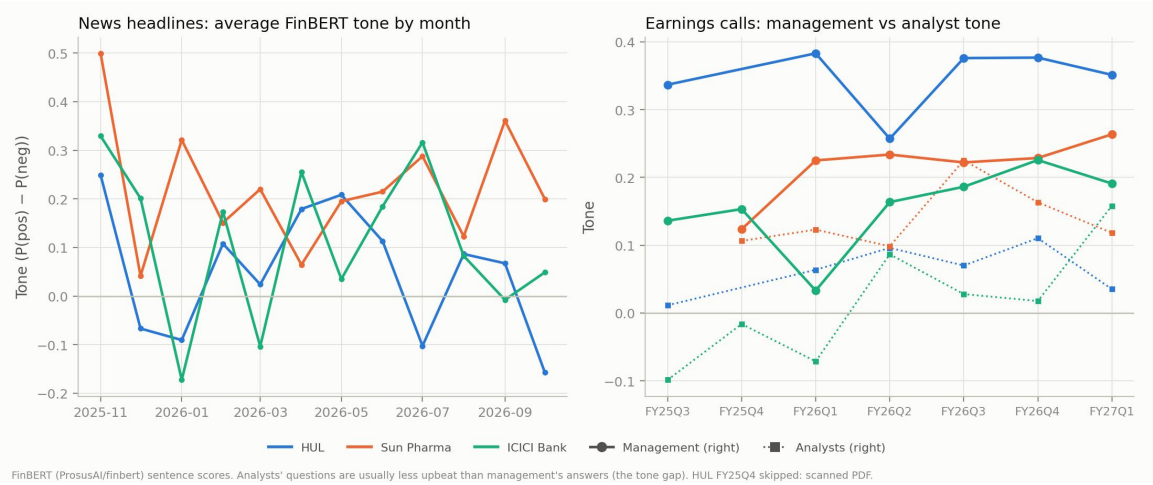


Figure A3. FinBERT tone: headlines by month (left) and earnings calls, management vs analysts (right).